Workshop 3

Maria Lucía Castillo García

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1. `docker-compose.yml`

Parte 1

```
casti@MLC MINGW64 ~/workshop3-docker-compose
$ cat > docker-compose.yml << 'EOF'
> services:
>   web:
>     image: nginx:alpine
>     ports:
>       - "8080:80"
EOF
```

```
services:
  web:
    image: nginx:alpine
    ports:
      - "8080:80"
```

Parte 2

```
casti@MLC MINGW64 ~/workshop3-docker-compose
$ notepad docker-compose.yml
```

```
services:
  web:
    image: nginx:alpine
    ports:
      - "8080:80"
    volumes:
      - ./html:/usr/share/nginx/html:ro
```

```
services:
  web:
    image: nginx:alpine
    ports:
      - "8080:80"
    volumes:
    -
./html:/usr/share/nginx/html:ro
```

Aclaración: la parte 3 quedó en el entregable `docker-compose.yml` de la raíz repo. Y la parte 4 en la carpeta de reto. Por eso no se encuentra necesario adjuntar la evidencia fotográfica.

2. Servicios funcionando correctamente.

Parte 4

```
casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ docker compose ps
NAME            IMAGE           COMMAND                  SERVICE   CREATED              STATUS              PORTS
reto-monitor-1  reto-monitor    "/bin/bash"              monitor   About a minute ago   Up About a minute
reto-web-1      reto-web        "/docker-entrypoint.…"   web       About a minute ago   Up About a minute   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
```

```
casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ docker compose exec web ls //logs
registro.log
```

```
casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ ls -la logs
total 5
drwxr-xr-x 1 casti 197609   0 Aug 30 05:47 ./
drwxr-xr-x 1 casti 197609   0 Aug 30 05:33 ../
-rw-r--r-- 1 casti 197609 212 Aug 30 05:47 registro.log

casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ cat logs/registro.log
===== Registro del 2026-08-30 10:47:26 =====
Usuario: root
Contenedor: a43f2ff788cb
Archivos en /logs: 1

===== Registro del 2026-08-30 10:47:45 =====
Usuario: root
Contenedor: a43f2ff788cb
Archivos en /logs: 1

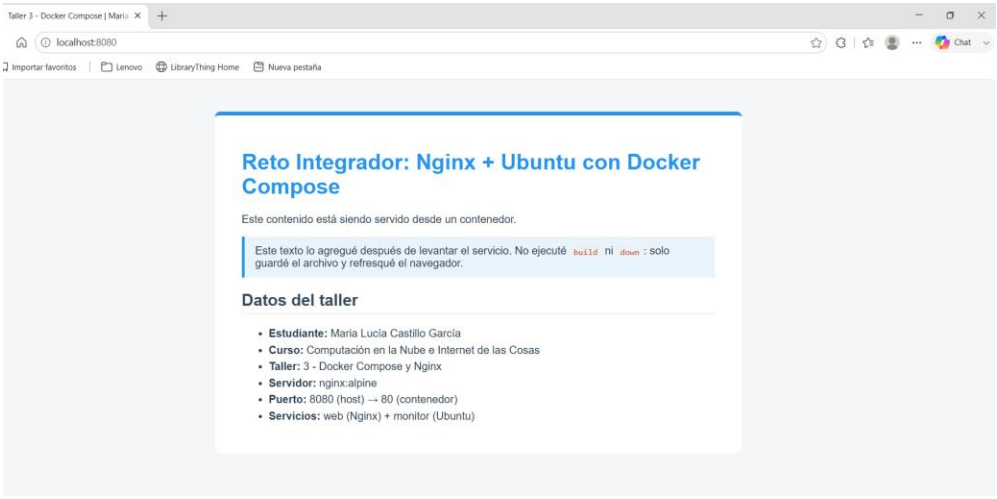

casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ docker compose ps
NAME             IMAGE          COMMAND                  SERVICE   CREATED         STATUS         PORTS
reto-monitor-1   reto-monitor   "/bin/bash"              monitor   7 minutes ago   Up 7 minutes
reto-web-1       reto-web       "/docker-entrypoint.…"   web       7 minutes ago   Up 7 minutes   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
```

3. Evidencia de `docker compose ps`.

```
casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ docker compose ps
NAME             IMAGE          COMMAND                  SERVICE   CREATED              STATUS              PORTS
reto-monitor-1   reto-monitor   "/bin/bash"              monitor   About a minute ago   Up About a minute
reto-web-1       reto-web       "/docker-entrypoint.…"   web       About a minute ago   Up About a minute   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
```

4. Evidencia de acceso a `Nginx`.

Parte 4



5. Respuestas a las preguntas de análisis.

Preguntas – Punto 2

- *¿Por qué el cambio aparece sin reconstruir la imagen?*

  - Porque con el volumen, el archivo nunca estuvo dentro de la imagen. Resulta que la línea `./html:/usr/share/nginx/html` conecta mi carpeta `html/`

con la carpeta que Nginx usa para servir páginas, así que las dos son la misma. Esto hizo que cuando editara el `index.html` en mi computador, Nginx ya estaba leyendo ese archivo cambiado (por lo que no había nada que reconstruir).

Lo comprobé editando el <h1>, refrescando el navegador y viendo el cambio sin ejecutar ningún comando de Docker.

❖ *¿Qué función cumple `:ro`?*

- Significa read only (solo lectura). Esto sirve porque prácticamente le dice a Docker que el contenedor puede leer los archivos de mi carpeta, pero no modificarlos ni borrarlos. Yo desde mi computador sí puedo editarlos con normalidad. Es una medida de seguridad porque si algo dentro del contenedor fallara o fuera atacado, no podría dañar los archivos de mi máquina.

❖ *¿Cuál es la diferencia entre el puerto `8080` y el `80`?*

- La diferencia es que son dos puertos en dos lugares distintos. El 80 es el puerto donde Nginx escucha dentro del contenedor, y siempre es ese porque es el estándar de la web.

Mientras que, el 8080 es el puerto de mi computador por donde yo entro a ese servicio.

Entonces, la línea "8080:80" construye el puente entre los dos: cuando abro localhost:8080, Docker redirige esa petición al puerto 80 del contenedor. Podría haber usado 3000 o 9090 en lugar de 8080 y funcionaría igual, siempre y cuando, el segundo número siga siendo el 80.

❖ *¿Qué parte representa el host y qué parte representa el contenedor?*

- Resulta ser que siempre sigue un determinado orden que es: primero el host, después el contenedor.

En "8080:80", el 8080 es mi computador (host) y el 80 es el contenedor. Y en `./html:/usr/share/nginx/html:ro` pasa lo mismo: `./html` es la carpeta en mi máquina y `/usr/share/nginx/html` es la ruta dentro del contenedor.

La regla es fácil de recordar porque va de afuera hacia adentro.

Pregunta análisis

- *Explique por qué Docker Compose resulta más conveniente que ejecutar manualmente varios `docker run` cuando un sistema comienza a tener múltiples servicios.*

  - Básicamente Docker Compose resulta más conveniente porque evita tener que escribir y recordar muchos comandos docker run. Resulta que cuando el sistema tiene varios servicios, como una aplicación, una base de datos y un servidor web, ejecutar cada contenedor manualmente se vuelve repetitivo y propenso a errores. Con Compose, toda la configuración queda en un solo archivo y basta con un comando para levantar o detener todo el sistema.

  Esto lo pude comprobar en la Parte 4, pues con un solo `docker compose up --build -d` se construyeron dos imágenes, se levantaron dos servicios, se creó la red que los comunica y se montó el volumen compartido entre ambos. Hacer lo mismo a mano habría requerido varios docker build, dos docker run con sus puertos y volúmenes, y crear la red aparte.

6. Anexo: Desarrollo del taller

```
casti@MLC MINGW64 ~/workshop2-docker-clase-asincronica/caso-c
$ cd ~

casti@MLC MINGW64 ~
$ mkdir -p ~/workshop3-docker-compose

casti@MLC MINGW64 ~
$ cd ~/workshop3-docker-compose

casti@MLC MINGW64 ~/workshop3-docker-compose
$ pwd
/c/Users/casti/workshop3-docker-compose
```

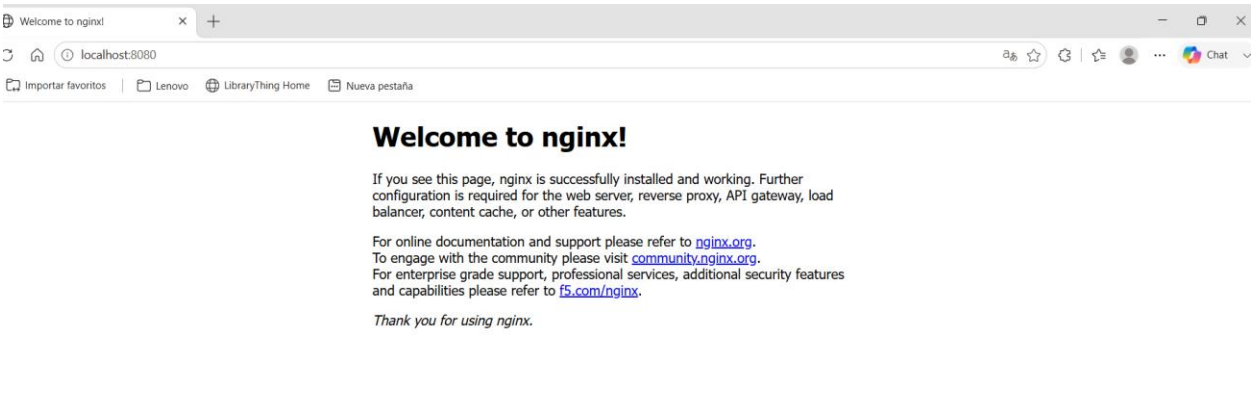
Parte 1. Primer servicio Nginx

```
casti@MLC MINGW64 ~/workshop3-docker-compose
$ cat > docker-compose.yml << 'EOF'
> services:
>   web:
>     image: nginx:alpine
      ports:
        - "8080:80"
EOF
```

```
casti@MLC MINGW64 ~/workshop3-docker-compose
$ cat docker-compose.yml
services:
  web:
    image: nginx:alpine
    ports:
      - "8080:80"
```

```
casti@MLC MINGW64 ~/workshop3-docker-compose
$ docker compose up -d
[+] up 13/13
 ✔ Image nginx:alpine                          Pulled
 ✔ Network workshop3-docker-compose_default    Created
 ✔ Container workshop3-docker-compose-web-1    Started
```

```
casti@MLC MINGW64 ~/workshop3-docker-compose
$ docker compose ps
NAME                              IMAGE          COMMAND                  SERVICE   CREATED          STATUS          PORTS
workshop3-docker-compose-web-1    nginx:alpine   "/docker-entrypoint.…"   web       25 seconds ago   Up 25 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
```




```
casti@MLC MINGW64 ~/workshop3-docker-compose
$ docker compose logs web
web-1  | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
web-1  | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
web-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
web-1  | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
web-1  | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
web-1  | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
web-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
web-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
web-1  | /docker-entrypoint.sh: Configuration complete; ready for start up
web-1  | 2026/08/30 09:01:48 [notice] 1#1: using the "epoll" event method
web-1  | 2026/08/30 09:01:48 [notice] 1#1: nginx/1.31.4
web-1  | 2026/08/30 09:01:48 [notice] 1#1: built by gcc 15.2.0 (Alpine 15.2.0)
web-1  | 2026/08/30 09:01:48 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
web-1  | 2026/08/30 09:01:48 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
web-1  | 2026/08/30 09:01:48 [notice] 1#1: start worker processes
web-1  | 2026/08/30 09:01:48 [notice] 1#1: start worker process 30
web-1  | 2026/08/30 09:01:48 [notice] 1#1: start worker process 31
web-1  | 2026/08/30 09:01:48 [notice] 1#1: start worker process 32
web-1  | 2026/08/30 09:01:48 [notice] 1#1: start worker process 33
web-1  | 2026/08/30 09:01:48 [notice] 1#1: start worker process 34
web-1  | 2026/08/30 09:01:48 [notice] 1#1: start worker process 35
web-1  | 2026/08/30 09:01:48 [notice] 1#1: start worker process 36
web-1  | 2026/08/30 09:01:48 [notice] 1#1: start worker process 37
web-1  | 2026/08/30 09:01:48 [notice] 1#1: start worker process 38
web-1  | 2026/08/30 09:01:48 [notice] 1#1: start worker process 39
web-1  | 2026/08/30 09:01:48 [notice] 1#1: start worker process 40
web-1  | 2026/08/30 09:01:48 [notice] 1#1: start worker process 41
web-1  | 172.18.0.1 - - [30/Aug/2026:09:03:03 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/152.0.0.0 Safari/537.36 Edg/152.0.0.0" "-"
web-1  | 172.18.0.1 - - [30/Aug/2026:09:03:03 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:8080/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/152.0.0.0 Safari/537.36 Edg/152.0.0.0" "-"
web-1  | 2026/08/30 09:03:03 [error] 30#30: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 172.18.0.1, server: localhost, request: "GET /favicon.ico HTTP/1.1", host: "localhost:8080", referrer: "http://localhost:8080/"
```

```
casti@MLC MINGW64 ~/workshop3-docker-compose
$ docker compose down
[+] down 2/2
 ✔ Container workshop3-docker-compose-web-1    Removed
 ✔ Network workshop3-docker-compose_default    Removed
```

Parte 2. Nginx con contenido propio

```
casti@MLC MINGW64 ~/workshop3-docker-compose
$ mkdir -p html

casti@MLC MINGW64 ~/workshop3-docker-compose
$ touch html/index.html

casti@MLC MINGW64 ~/workshop3-docker-compose
$ ls -la html
total 0
drwxr-xr-x 1 casti 197609 0 Aug 30 04:10 ./
drwxr-xr-x 1 casti 197609 0 Aug 30 04:10 ../
-rw-r--r-- 1 casti 197609 0 Aug 30 04:10 index.html

casti@MLC MINGW64 ~/workshop3-docker-compose
$ notepad html/index.html
```

```
<!DOCTYPE html>
<html lang="es">
<head>
    <meta charset="UTF-8">
    <title>Taller 3 - Docker Compose | Maria Lucía Castillo García</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f4f6f8;
            color: #2c3e50;
            padding: 40px;
        }
        .contenedor {
            max-width: 700px;
            margin: 0 auto;
            background-color: #ffffff;
            border-top: 6px solid #2496ed;
            border-radius: 8px;
            padding: 30px 40px;
        }
        h1 {
            color: #2496ed;
        }
        h2 {
            color: #34495e;
            border-bottom: 1px solid #dde3e8;
            padding-bottom: 6px;
        }
        code {
            background-color: #eef2f5;
            color: #c0392b;
            padding: 2px 6px;
            border-radius: 3px;
        }
        li {
            margin-bottom: 6px;
        }
        .nota {
            background-color: #eaf4fd;
            border-left: 4px solid #2496ed;
```

```
        .nota {
            background-color: #eaf4fd;
            border-left: 4px solid #2496ed;
            padding: 12px 16px;
        }
    </style>
</head>
<body>
    <div class="contenedor">
        <h1>Hola desde Nginx y Docker Compose</h1>
        <p>Este contenido está siendo servido desde un contenedor.</p>

        <h2>Datos del taller</h2>
        <ul>
            <li><strong>Estudiante:</strong> Maria Lucía Castillo García</li>
            <li><strong>Curso:</strong> Computación en la Nube e Internet de las Cosas</li>
            <li><strong>Taller:</strong> 3 - Docker Compose y Nginx</li>
            <li><strong>Servidor:</strong> nginx:alpine</li>
            <li><strong>Puerto:</strong> 8080 (host) &rarr; 80 (contenedor)</li>
        </ul>

        <p class="nota">Esta página no está dentro de la imagen: se monta como
        volumen desde la carpeta <code>html/</code> de mi computador.</p>
    </div>
</body>
</html>
```

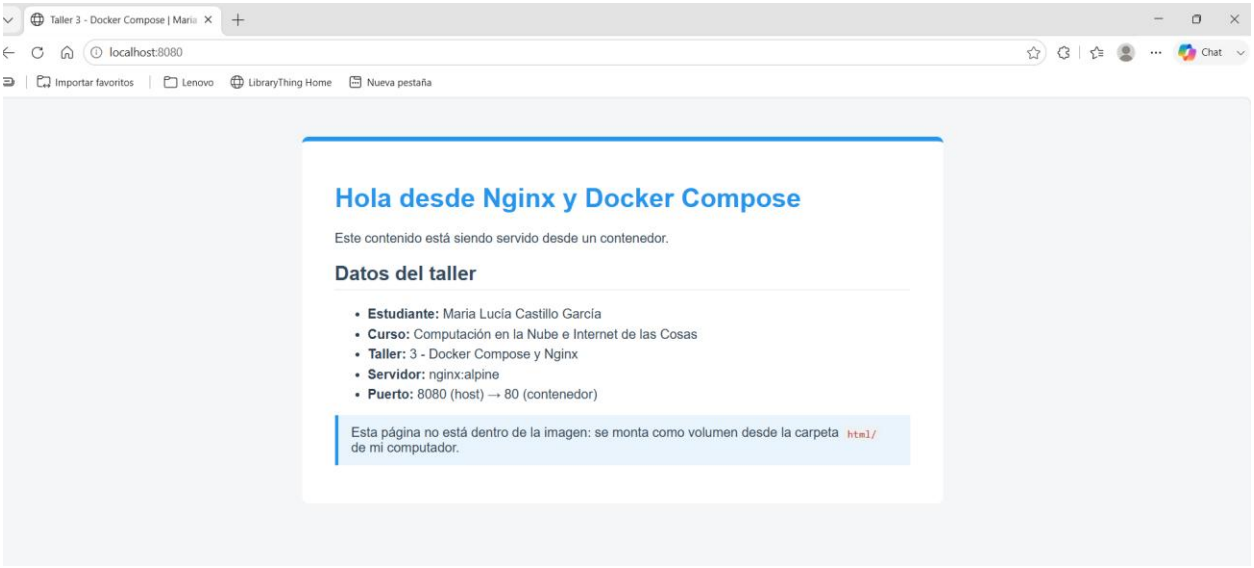
```
casti@MLC MINGW64 ~/workshop3-docker-compose
$ notepad docker-compose.yml
```

```
services:
  web:
    image: nginx:alpine
    ports:
      - "8080:80"
    volumes:
      - ./html:/usr/share/nginx/html:ro
```

```
casti@MLC MINGW64 ~/workshop3-docker-compose
$ cat docker-compose.yml
services:
  web:
    image: nginx:alpine
    ports:
      - "8080:80"
    volumes:
      - ./html:/usr/share/nginx/html:ro
```

```
casti@MLC MINGW64 ~/workshop3-docker-compose
$ docker compose up -d
[+] up 2/2
 ✔ Network workshop3-docker-compose_default Created
 ✔ Container workshop3-docker-compose-web-1 Started

casti@MLC MINGW64 ~/workshop3-docker-compose
$ docker compose ps
NAME                              IMAGE          COMMAND                  SERVICE   CREATED          STATUS          PORTS
workshop3-docker-compose-web-1   nginx:alpine   "/docker-entrypoint.…"   web       10 seconds ago   Up 10 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
```



Experimento: modificar el archivo sin reconstruir

```
casti@MLC MINGW64 ~/workshop3-docker-compose
$ notepad html/index.html
```

```
        }
        .nota {
            background-color: #eaf4fd;
            border-left: 4px solid #2496ed;
            padding: 12px 16px;
        }
    </style>
</head>
<body>
    <div class="contenedor">
        <h1>Página modificada SIN reconstruir la imagen</h1>
        <p>Este contenido está siendo servido desde un contenedor.</p>
        <p class="nota">Este texto lo agregué después de levantar el servicio.
        No ejecuté <code>build</code> ni <code>down</code>: solo guardé el
        archivo y refresqué el navegador.</p>

        <h2>Datos del taller</h2>
        <ul>
            <li><strong>Estudiante:</strong> Maria Lucía Castillo García</li>
            <li><strong>Curso:</strong> Computación en la Nube e Internet de las Cosas</li>
            <li><strong>Taller:</strong> 3 - Docker Compose y Nginx</li>
            <li><strong>Servidor:</strong> nginx:alpine</li>
            <li><strong>Puerto:</strong> 8080 (host) &rarr; 80 (contenedor)</li>
        </ul>

        <p class="nota">Esta página no está dentro de la imagen: se monta como
```



→ En ambas capturas no se ejecutó ningún comando de Docker

```
casti@MLC MINGW64 ~/workshop3-docker-compose
$ docker compose down
[+] down 2/2
 ✔ Container workshop3-docker-compose-web-1 Removed
 ✔ Network workshop3-docker-compose_default Removed
```

Parte 3. Compose utilizando un Dockerfile

```
casti@MLC MINGW64 ~/workshop3-docker-compose
$ touch Dockerfile

casti@MLC MINGW64 ~/workshop3-docker-compose
$ notepad Dockerfile
```

```
FROM nginx:alpine
COPY html/ /usr/share/nginx/html/
```

```
casti@MLC MINGW64 ~/workshop3-docker-compose
$ cat Dockerfile
FROM nginx:alpine
COPY html/ /usr/share/nginx/html/
casti@MLC MINGW64 ~/workshop3-docker-compose
```

```
casti@MLC MINGW64 ~/workshop3-docker-compose
$ notepad docker-compose.yml
```

```
services:
  web:
    build: .
    ports:
      - "8080:80"
```

```
casti@MLC MINGW64 ~/workshop3-docker-compose
$ cat docker-compose.yml
services:
  web:
    build: .
    ports:
      - "8080:80"
```

```
casti@MLC MINGW64 ~/workshop3-docker-compose
$ docker compose up --build -d
#1 [internal] load local bake definitions
#1 reading from stdin 559B 0.0s done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 89B 0.0s done
#2 DONE 0.0s

#3 [internal] load metadata for docker.io/library/nginx:alpine
#3 DONE 0.1s

#4 [internal] load .dockerignore
#4 transferring context: 2B done
#4 DONE 0.0s

#5 [internal] load build context
#5 transferring context: 2.24kB done
#5 DONE 0.0s

#6 [1/2] FROM docker.io/library/nginx:alpine@sha256:db35bfc6b2951e7f8a72db5db120288c127ffaeeb4a6d4b95a26fead017d5913
#6 resolve docker.io/library/nginx:alpine@sha256:db35bfc6b2951e7f8a72db5db120288c127ffaeeb4a6d4b95a26fead017d5913 0.0s done
#6 DONE 0.2s

#7 [2/2] COPY html/ /usr/share/nginx/html/
#7 DONE 0.0s

#8 exporting to image
#8 exporting layers 0.1s done
#8 exporting manifest sha256:914d1eced36952f06ce76af0f4b48956fe6014e88c201801dff5d404063fac82 0.0s done
#8 exporting config sha256:c583ddc6ff93edba646cfcc56995c58d60c8a6d73cfd4edf1905214769f3b5c2 0.0s done
#8 exporting attestation manifest sha256:8047bce5c0c39d5e351b919799f65a13c4d315c25e7bfef290f72fed02ec98d1 0.0s done
#8 exporting manifest list sha256:b38b8152078ad9f86e0e63cb96c83729a8098809d70a4eb28def1d0df4dee5f3 0.0s done
#8 naming to docker.io/library/workshop3-docker-compose-web:latest
#8 naming to docker.io/library/workshop3-docker-compose-web:latest done
#8 unpacking to docker.io/library/workshop3-docker-compose-web:latest 0.1s done
#8 DONE 0.3s

#9 resolving provenance for metadata file
#9 DONE 0.0s
[+] up 3/3
 ✔ Image workshop3-docker-compose-web       Built
 ✔ Network workshop3-docker-compose_default Created
 ✔ Container workshop3-docker-compose-web-1 Started
```

```
casti@MLC MINGW64 ~/workshop3-docker-compose
$ docker images

IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
docker/welcome-to-docker:latest      c4d56c24da4f       22.2MB         6.03MB    U
ejemplo-cmd:latest                   a30a7160aef4        157MB         41.6MB
ejemplo-combinado:latest             7f54bcab2e11        157MB         41.6MB
ejemplo-entrypoint:latest            603694ffee4c        157MB         41.6MB
japeto/pujgcc:latest                 9b3d7bbf4103       2.18GB          448MB    U
japeto/so-tools:v64                  7e990f123c8d         2GB            506MB    U
nginx:alpine                         db35bfc6b295       94.2MB         27.2MB
ubuntu-bash-lab:latest               c4ff932c0a29        162MB           43MB    U
ubuntu:latest                        2260313b31c8        160MB         45.3MB
workshop3-docker-compose-web:latest  b38b8152078a       93.4MB         26.3MB    U
```



ACLARACIÓN: Aunque en la página dice que la página no está dentro de la imagen, de hecho, en este caso sí. Solo que dejé el mismo formato de todo en la página, entonces se mantuvo el del punto anterior donde tenía ese aviso.

Experimento inverso: modificar el archivo sin reconstruir

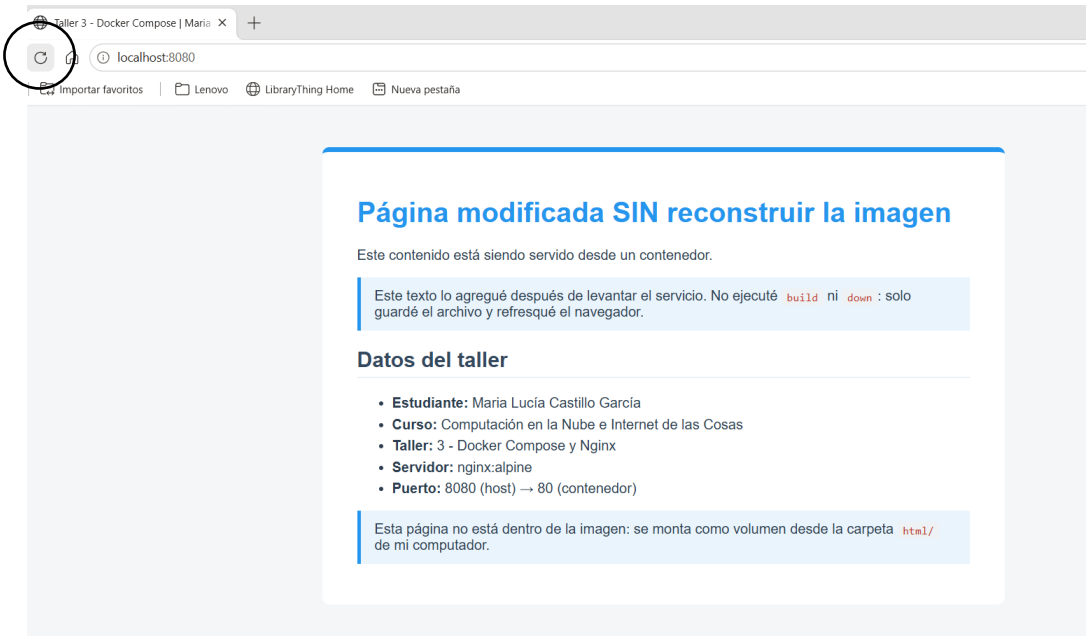
```
casti@MLC MINGW64 ~/workshop3-docker-compose
$ notepad html/index.html
```

```
        }
    </style>
</head>
<body>
    <div class="contenedor">
        <h1>Este cambiecito NO se ve hasta reconstruir</h1>
        <p>Este contenido está siendo servido desde un contenedor.</p>
        <p class="nota">Este texto lo agregué después de levantar el servicio.
        No ejecuté <code>build</code> ni <code>down</code>: solo guardé el
        archivo y refresqué el navegador.</p>

        <h2>Datos del taller</h2>
        <ul>
            <li><strong>Estudiante:</strong> Maria Lucía Castillo García</li>
            <li><strong>Curso:</strong> Computación en la Nube e Internet de las Cosas</li>
            <li><strong>Taller:</strong> 3 - Docker Compose y Nginx</li>
            <li><strong>Servidor:</strong> nginx:alpine</li>
```

```
            <li><strong>Puerto:</strong> 8080 (host) &rarr; 80 (contenedor)</li>
        </ul>

        <p class="nota">Esta página no está dentro de la imagen: se monta como
        volumen desde la carpeta <code>html/</code> de mi computador.</p>
```



```
$ notepad html/index.html

casti@MLC MINGW64 ~/workshop3-docker-compose
$ docker compose up --build -d
#1 [internal] load local bake definitions
#1 reading from stdin 559B 0.0s done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile:
#2 transferring dockerfile: 89B done
#2 DONE 0.0s

#3 [internal] load metadata for docker.io/library/nginx:alpine
#3 DONE 0.0s

#4 [internal] load .dockerignore
#4 transferring context: 2B done
#4 DONE 0.0s

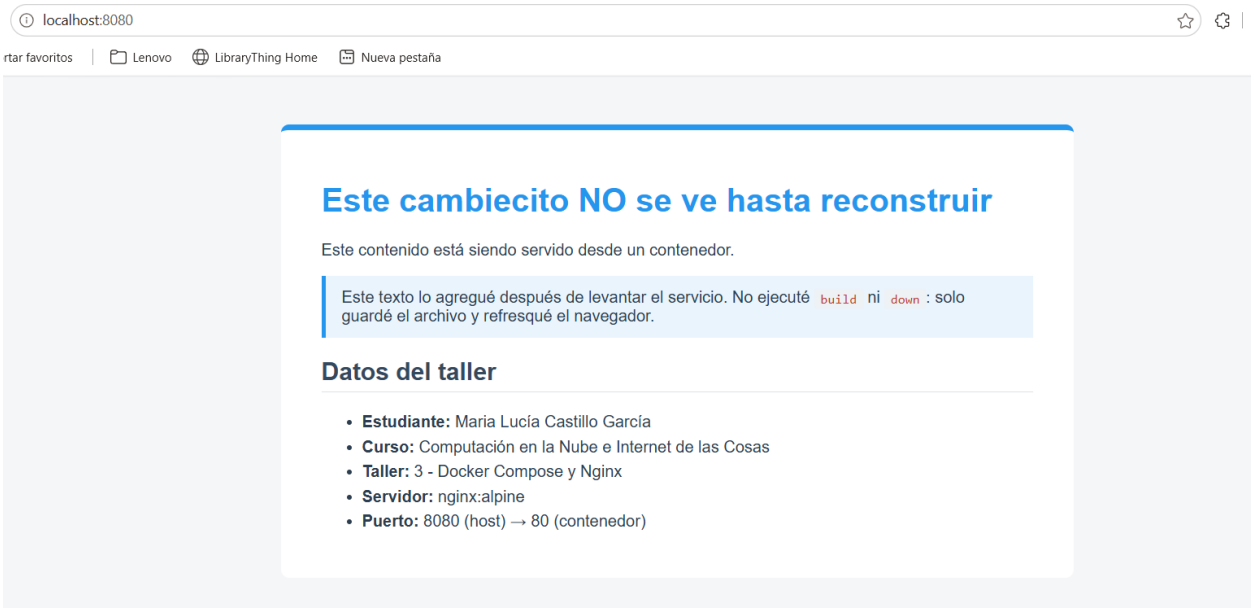
#5 [internal] load build context
#5 transferring context: 2.08kB done
#5 DONE 0.0s

#6 [1/2] FROM docker.io/library/nginx:alpine@sha256:db35bfc6b2951e7f8a72db5db120288c127ffaeeb4a6d4b95a26fead017d5913
#6 resolve docker.io/library/nginx:alpine@sha256:db35bfc6b2951e7f8a72db5db120288c127ffaeeb4a6d4b95a26fead017d5913 0.0s done
#6 CACHED

#7 [2/2] COPY html/ /usr/share/nginx/html/
#7 DONE 0.0s

#8 exporting to image
#8 exporting layers 0.1s done
#8 exporting manifest sha256:5908eb7bba9e5dff9e895a3b085468da401157b36557ba5396982461b2b8f593 0.0s done
#8 exporting config sha256:288f130c818a6ffc3522c350f7bf118b0aa1769518eba6b09c34ae9ed25604f7 0.0s done
#8 exporting attestation manifest sha256:f9311ac175a0a2fee6f617e9ffb6a146f940f2a47e74d00d0f16fd1baa539e12 0.0s done
#8 exporting manifest list sha256:747e028671d5dc259b42144a984acc75bbabfc26911d2e26144d7bc4741a2468
#8 exporting manifest list sha256:747e028671d5dc259b42144a984acc75bbabfc26911d2e26144d7bc4741a2468 0.0s done
#8 naming to docker.io/library/workshop3-docker-compose-web:latest done
#8 unpacking to docker.io/library/workshop3-docker-compose-web:latest 0.0s done
#8 DONE 0.3s

#9 resolving provenance for metadata file
#9 DONE 0.0s
[+] up 2/2
 ✔ Image workshop3-docker-compose-web       Built
 ✔ Container workshop3-docker-compose-web-1 Started
```




```
casti@MLC MINGW64 ~/workshop3-docker-compose
$ docker compose down
[+] down 2/2
 ✔ Container workshop3-docker-compose-web-1 Removed
 ✔ Network workshop3-docker-compose_default Removed
```

Parte 4. Reto integrador

1. Estructura del proyecto

```
casti@MLC MINGW64 ~/workshop3-docker-compose
$ mkdir -p reto/ubuntu reto/logs
```

```
casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ mkdir -p html

casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ cp ../html/index.html html/

casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ ls -la
total 4
drwxr-xr-x 1 casti 197609 0 Aug 30 05:08 ./
drwxr-xr-x 1 casti 197609 0 Aug 30 05:06 ../
drwxr-xr-x 1 casti 197609 0 Aug 30 05:08 html/
drwxr-xr-x 1 casti 197609 0 Aug 30 05:06 logs/
drwxr-xr-x 1 casti 197609 0 Aug 30 05:06 ubuntu/
```

```
casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ touch Dockerfile

casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ notepad Dockerfile
```

```
FROM nginx:alpine
COPY html/ /usr/share/nginx/html/
```

```
casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ cat Dockerfile
FROM nginx:alpine
COPY html/ /usr/share/nginx/html/
casti@MLC MINGW64 ~/workshop3-docker-compose/reto
```

2. Los scripts monitor.sh y analiza.sh

```
casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ nano ubuntu/monitor.sh

casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ nano ubuntu/analiza.sh

casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ cat ubuntu/monitor.sh
#!/bin/bash
FECHA=$(date '+%Y-%m-%d %H:%M:%S')

echo "===== Registro del $FECHA =====" >> /logs/registro.log
echo "Usuario: $(whoami)" >> /logs/registro.log
echo "Contenedor: $(hostname)" >> /logs/registro.log
echo "Archivos en /logs: $(ls /logs | wc -l)" >> /logs/registro.log
echo "" >> /logs/registro.log

echo "Registro guardado en /logs/registro.log"

casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ cat ubuntu/analiza.sh
#!/bin/bash
ARCHIVO="/logs/registro.log"

if [ -f "$ARCHIVO" ]
then
    echo "Analizando $ARCHIVO"
    echo "Lineas totales: $(wc -l < $ARCHIVO)"
    echo "Registros realizados: $(grep -c 'Registro del' $ARCHIVO)"
    echo "Ultimo registro:"
    grep 'Registro del' $ARCHIVO | tail -1
else
    echo "No existe $ARCHIVO. Ejecute primero monitor.sh"
fi
```

3. **Dockerfile del servicio Ubuntu**

```
casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ nano ubuntu/Dockerfile

casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ cat ubuntu/Dockerfile
FROM ubuntu:latest

WORKDIR /scripts

COPY monitor.sh analiza.sh /scripts/

RUN chmod +x /scripts/monitor.sh /scripts/analiza.sh
```

4. **Los dos servicios en uno solo `docker-compose.yml`, construcción y arranque simultáneo; y Nginx disponible en localhost:8080**

```
casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ notepad docker-compose.yml

casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ cat docker-compose.yml
services:
  web:
    build: .
    ports:
      - "8080:80"
    volumes:
      - ./logs:/logs

  monitor:
    build: ./ubuntu
    volumes:
      - ./logs:/logs
    tty: true
casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$
```

→ ACLARACIÓN:

`tty:` true mantiene una terminal abierta en el contenedor de Ubuntu. Es la misma opción -t de docker run -it. Sin ella, el contenedor ejecutaría su bash, no encontraría a nadie escribiendo y se apagaría de inmediato, impidiendo demostrar los dos servicios corriendo al tiempo.

```
casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ cd ~/workshop3-docker-compose/

casti@MLC MINGW64 ~/workshop3-docker-compose
$ docker compose down

casti@MLC MINGW64 ~/workshop3-docker-compose
$ cd reto
```

```
#15 [monitor 3/4] COPY monitor.sh analiza.sh /scripts/
#15 DONE 0.1s

#14 [web] exporting to image
#14 exporting attestation manifest sha256:88667d731af6ca84a4a2cd597e9b600771O1e08fa9912be23980cb815ea31f25 0.1s done
#14 exporting manifest list sha256:480ebff373bf49e6646cb531d1c6581ac1eac1784bced760f46f059d3ebc046d 0.0s done
#14 naming to docker.io/library/reto-web:latest
#14 naming to docker.io/library/reto-web:latest done
#14 unpacking to docker.io/library/reto-web:latest 0.0s done
#14 DONE 0.3s

#16 [web] resolving provenance for metadata file
#16 DONE 0.0s

#17 [monitor 4/4] RUN chmod +x /scripts/monitor.sh /scripts/analiza.sh
#17 DONE 0.4s

#18 [monitor] exporting to image
#18 exporting layers 0.2s done
#18 exporting manifest sha256:efac4ceaac1affbcdd90d2df277455fa20478a52ac33aa1990484d28cb514003 0.0s done
#18 exporting config sha256:183f55ecb18ec1b9b33dd4c1d1d8619deb06ed6a0a8e6ef883304a55c0512070 0.0s done
#18 exporting attestation manifest sha256:712e00e0d3aadb151dfb957e5a2d2d75612254cc6eed45a5753415c5cffdee0a 0.0s done
#18 exporting manifest list sha256:d581826505ea74ea67b1cb94b1dfccab7097fe21a0a023ebf9caf64a80a4aa8e 0.0s done
#18 naming to docker.io/library/reto-monitor:latest
#18 naming to docker.io/library/reto-monitor:latest done
#18 unpacking to docker.io/library/reto-monitor:latest 0.1s done
#18 DONE 0.5s

#19 [monitor] resolving provenance for metadata file
#19 DONE 0.0s
[+] up 5/5
 ✔ Image reto-monitor       Built
 ✔ Image reto-web           Built
 ✔ Network reto_default     Created
 ✔ Container reto-web-1     Started
 ✔ Container reto-monitor-1 Started
```

```
asti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ docker compose up --build -d
#1 [internal] load local bake definitions
#1 reading from stdin 1.01kB 0.0s done
#1 DONE 0.0s

#2 [monitor internal] load build definition from Dockerfile
#2 transferring dockerfile: 170B done
#2 DONE 0.0s

#3 [web internal] load build definition from Dockerfile
#3 transferring dockerfile: 89B done
#3 DONE 0.0s

#4 [web internal] load metadata for docker.io/library/nginx:alpine
#4 DONE 0.1s

#5 [monitor internal] load metadata for docker.io/library/ubuntu:latest
#5 DONE 0.1s

#6 [web internal] load .dockerignore
#6 transferring context: 2B done
#6 DONE 0.0s

#7 [monitor internal] load .dockerignore
#7 transferring context: 2B done
#7 DONE 0.0s

#8 [web internal] load build context
#8 transferring context: 2.17kB done
#8 DONE 0.0s

#9 [monitor 1/4] FROM docker.io/library/ubuntu:latest@sha256:2260313b31c8c011cd2eebe728008efac1b3982be73eb71348ea2648d2c0e09b
#9 resolve docker.io/library/ubuntu:latest@sha256:2260313b31c8c011cd2eebe728008efac1b3982be73eb71348ea2648d2c0e09b 0.0s done
#9 DONE 0.0s

#10 [web 1/2] FROM docker.io/library/nginx:alpine@sha256:db35bfc6b2951e7f8a72db5db120288c127ffaeeb4a6d4b95a26fead017d5913
#10 resolve docker.io/library/nginx:alpine@sha256:db35bfc6b2951e7f8a72db5db120288c127ffaeeb4a6d4b95a26fead017d5913 0.0s done
#10 CACHED

#11 [monitor internal] load build context
#11 transferring context: 62B done
#11 DONE 0.0s

#12 [monitor 2/4] WORKDIR /scripts
#12 CACHED

#13 [monitor 3/4] COPY monitor.sh analiza.sh /scripts/
#13 CACHED

#14 [monitor 4/4] RUN chmod +x /scripts/monitor.sh /scripts/analiza.sh
#14 CACHED

#15 [web 2/2] COPY html/ /usr/share/nginx/html/
#15 DONE 0.1s

#16 [monitor] exporting to image
#16 exporting layers done
#16 exporting manifest sha256:efac4ceaac1affbcdd90d2df277455fa20478a52ac33aa1990484d28cb514003 done
#16 exporting config sha256:183f55ecb18ec1b9b33dd4c1d1d8619deb06ed6a0a8e6ef883304a55c0512070 done
#16 exporting attestation manifest sha256:fc08f5c0885266a3f0f99e0475c51f489d371ebdb5133867f70e2d88e10e73a5 0.1s done
#16 exporting manifest list sha256:10a354bd55d5b6a83405b09dce6a6596b1fd5a3fbcf3173f50bb294f554563fc
#16 exporting manifest list sha256:10a354bd55d5b6a83405b09dce6a6596b1fd5a3fbcf3173f50bb294f554563fc 0.0s done
#16 naming to docker.io/library/reto-monitor:latest done
#16 unpacking to docker.io/library/reto-monitor:latest 0.0s done
#16 DONE 0.2s
```

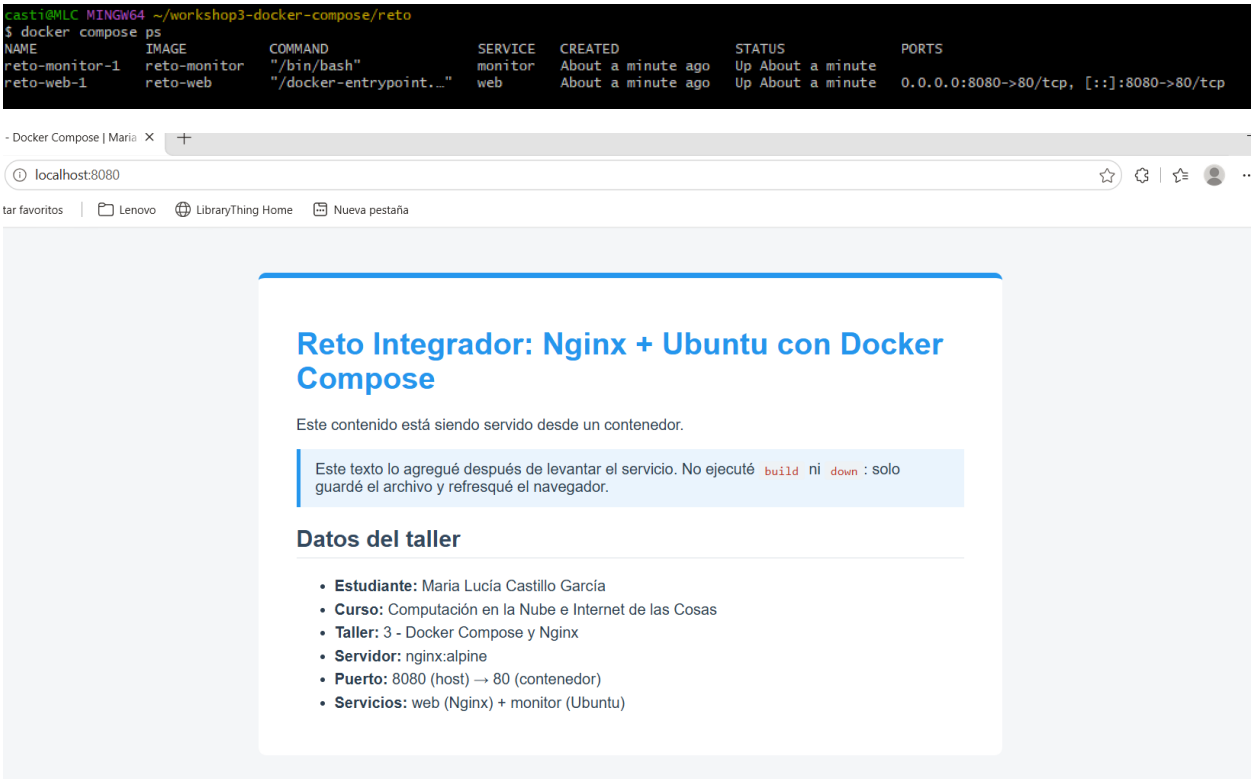
```
#16 exporting manifest sha256:efac4ceaac1affbcdd90d2df277455fa20478a52ac33aa1990484d28cb514003 done
#16 exporting config sha256:183f55ecb18ec1b9b33dd4c1d1d8619deb06ed6a0a8e6ef883304a55c0512070 done
#16 exporting attestation manifest sha256:fc08f5c0885266a3f0f99e0475c51f489d371ebdb5133867f70e2d88e10e73a5 0.1s done
#16 exporting manifest list sha256:10a354bd55d5b6a83405b09dce6a6596b1fd5a3fbcf3173f50bb294f554563fc
#16 exporting manifest list sha256:10a354bd55d5b6a83405b09dce6a6596b1fd5a3fbcf3173f50bb294f554563fc 0.0s done
#16 naming to docker.io/library/reto-monitor:latest done
#16 unpacking to docker.io/library/reto-monitor:latest 0.0s done
#16 DONE 0.2s

#17 [web] exporting to image
#17 exporting layers 0.3s done
#17 exporting manifest sha256:67d1150d4db93719d9cda803ad7089efd6bed3822a471dd0d3df55d852a4dc9a 0.0s done
#17 exporting config sha256:7903f363c445cbdc610011554f584862c8d1884805a6e20ac5426707330c56f23 0.0s done
#17 exporting attestation manifest sha256:9d9c9c01c2d8bf862d29fafc679d24733eb582a10a36d1d8fedce8176112d8c3
#17 exporting attestation manifest sha256:9d9c9c01c2d8bf862d29fafc679d24733eb582a10a36d1d8fedce8176112d8c3 0.0s done
#17 exporting manifest list sha256:939fed0e67000f85049db1f498dc91274938969d1d92792353732c2efa702fef 0.0s done
#17 naming to docker.io/library/reto-web:latest done
#17 unpacking to docker.io/library/reto-web:latest 0.0s done
#17 DONE 0.5s

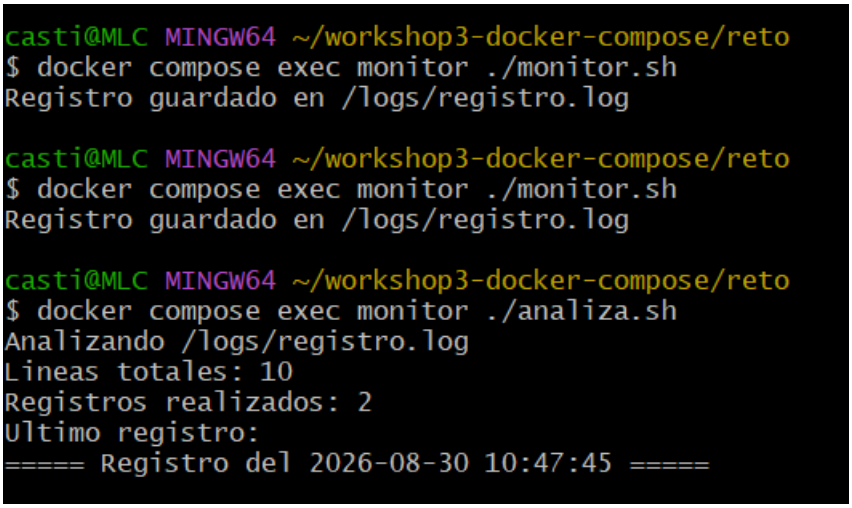
#18 [monitor] resolving provenance for metadata file
#18 DONE 0.0s

#19 [web] resolving provenance for metadata file
#19 DONE 0.0s
[+] up 4/4
 ✔ Image reto-monitor       Built
 ✔ Image reto-web           Built
 ✔ Container reto-monitor-1 Started
 ✔ Container reto-web-1     Started
```

5. Ejecución de los scripts dentro del contenedor



6. Verificación del volumen compartido entre servicios



```
casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ docker compose exec web ls -la /logs
ls: C:/Program Files/Git/logs: No such file or directory

What's next:
    Debug this Compose error with Gordon → docker ai "help me fix this compose error"

casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ docker compose exec web ls //logs
registro.log
```

→ Git bash produjo la ruta /logs al formato de Windows antes de enviarla al contenedor. Con //logs se evita esa conversión.

```
casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ ls -la logs
total 5
drwxr-xr-x 1 casti 197609   0 Aug 30 05:47 ./
drwxr-xr-x 1 casti 197609   0 Aug 30 05:33 ../
-rw-r--r-- 1 casti 197609 212 Aug 30 05:47 registro.log

casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ cat logs/registro.log
===== Registro del 2026-08-30 10:47:26 =====
Usuario: root
Contenedor: a43f2ff788cb
Archivos en /logs: 1

===== Registro del 2026-08-30 10:47:45 =====
Usuario: root
Contenedor: a43f2ff788cb
Archivos en /logs: 1


casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ docker compose ps
NAME            IMAGE          COMMAND                  SERVICE   CREATED         STATUS         PORTS
reto-monitor-1  reto-monitor   "/bin/bash"              monitor   7 minutes ago   Up 7 minutes
reto-web-1      reto-web       "/docker-entrypoint.…"   web       7 minutes ago   Up 7 minutes   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
```

```
casti@MLC MINGW64 ~/workshop3-docker-compose/reto
$ docker compose down
[+] down 3/3
 ✔ Container reto-web-1      Removed
 ✔ Container reto-monitor-1  Removed
 ✔ Network reto_default      Removed
```